

Leonardo Corso

DBGuard

User manual

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What is DBGuard?

DBGuard (Database Guard) is an application that periodically checks the state of your databases and sends notifications when specific conditions are met. It is useful for catching bad or unexpected data that can cause applications to crash, or for alerting a team when data meets any criteria that warrants attention.

DBGuard supports multiple database engines including SQL Server, PostgreSQL, SQLite, and MySQL.

Monitoring is performed through Guards — each Guard runs a user-defined SQL query on a periodic schedule and compares the result to a user-defined expression. Since the core logic and business rules are written in SQL, the application remains decoupled and simple to configure.

When a Guard's expression evaluates to true, it can trigger zero or more notifications via SMTP (email) or HTTP (webhooks to other applications or APIs). DBGuard also includes a built-in SPA web application for configuring, testing, and auditing Guards and their execution history.

Getting started

Guards

Understanding a guard

A guard is a single query against a single database that evaluates the result to a user defined expression. The guard is created and run at specific intervals as some guards may only need to run once a week, while others multiple times an hour. A guard can be in one of four states:

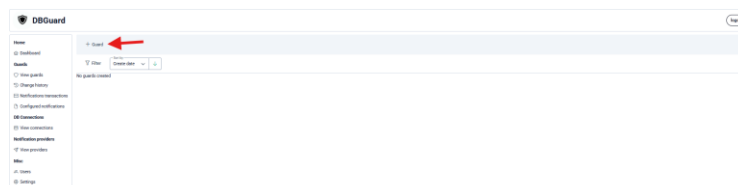
- **Unknown:** The initial state of the guard. Indicates the scan has not yet run.
- **Error:** The guard ran into one of various errors such as it could not connect to the database, the query is invalid, the result column it's defined to look at does not exist in the result set, or any other error which can occur when a query is ran from a local or remote application against a database.
- **Triggered:** The user defined query and expression are evaluated as true.
- **Clear:** The user defined query and expression is evaluated as false.

Guard database connection

A database connection is used by the guard to connect to a database. The database connection can be reused by many guards. If the database engine requires a user login, the database user should have only read permission to the database to ensure best security practices. The credentials are encrypted and stored in the PostgreSQL application database using an IDDataProtector. Some database engines such as SQLite do not use logins so it is not required.

Create a guard

1. To create a guard, go to “/guards/view-all” or select “View guards” from the sidebar menu.
2. Select the “+Guard” button on the toolbar:



3. You will then see the dialog to create the guard:

Create guard



Is active



Guard name

Guard description

Trigger query

Required

Trigger column

Required

Trigger Operator



Required

Trigger value

Required

Database connection



Required

Run period (minutes)

5



Required

Notifications



No notifications added



Notification provider



Notify on clear



Notify on error



Notify on trigger



Validate guard: ☒

Add

4. Here is what each field does:

- **Is active:** Boolean that determines if the guard is active or paused and should be run by the monitoring service.
- **Guard name:** Optional user-friendly name.
- **Guard description:** Optional description of the guard.
- **Trigger query:** The SQL query for the guard. The query needs to return a single row. The reason for this is that all the logic should be implemented in the query and we only need a single record so we can look at the trigger column to evaluate the expression.
- **Trigger column:** This is the name of the column in our queries result record that is used in the expression. Its value should be a 32-bit integer as it will be used in the expression to evaluate the guard state.
- **Trigger operator:** The operator for the expression. Supports equals, not equals, greater than, less than, greater than or equal to, less than or equal to.
- **Trigger value:** A 32-bit integer value for our expression.
- **Database connection:** A connection and login if required to our database.
- **Run period:** How often in minutes the guard will be run.
- **Notifications:** The notifications for the guard.
- **Notify on clear, error, trigger:** Whether the guard should notify using its configured notifications when it goes into the clear, error, or triggered state respectively.
- **Validate guard:** Whether to validate the guard before creation to ensure it can connect to the database, the query is valid and trigger column exists in the result set. Will return an error with failure if it is invalid.

5. Click add.

Example guard

For example, let's say we want to be notified once a day if there are any customers in our database who have a debt that is more than \$10.50. This is a good example to show that although the guard trigger column value is a 32-bit integer, it can support monitoring decimals, dates, strings and many other data types since the logic is stored in the SQL query, and the application simply checks a trigger value.

1. Our trigger query may look something like this:

```
SELECT COUNT(*) as HighDebt
```

```
FROM customer_debt
```

```
WHERE customer_debt.dollars > 10.5;
```

2. The column we want to look at is called "*HighDebt*" so the trigger column will be "HighDebt".
3. Since the query uses ">" our trigger operator will be greater than or equal to.
4. Lastly the trigger value will be 1 since we want to be notified if any exist.
5. The expression will then read as, trigger guard when "HighDebt" column in query result is greater than 1.
6. Since we want it to run once a day, we will set the run period in minutes to 1440, so it runs every 24 hours.

Create guard



Is active



Guard name

High Debt Example

Guard description

Demo guard to showcase a high debt

Trigger query

```
SELECT COUNT(*) AS HighDebt
FROM customer_debt
WHERE customer_debt.debt_dollars > 10.5;
```

Required

Trigger column

HighDebt

Required

Trigger Operator

Equals

Required

Trigger value

1

Required

Database connection

- Database type: PostgreSQL
- Endpoint: [REDACTED]
- Database name: [REDACTED]
- Username: [REDACTED]



Required

Run period (minutes)

1,440

Required

Notifications



No notifications added



Notification provider



Notify on clear



Notify on error



Notify on trigger



Validate guard: ☒

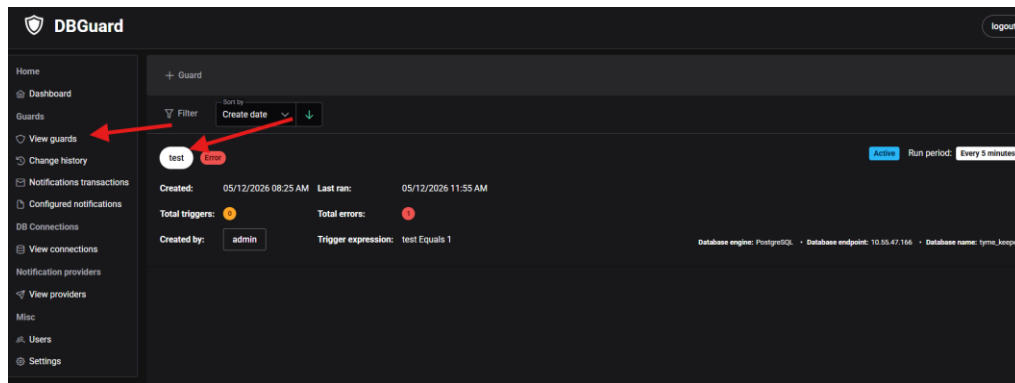
Add

View guards

You can view the created guards by selecting the “View guards” navigation on the sidebar or navigating to “/guards/view-all”.

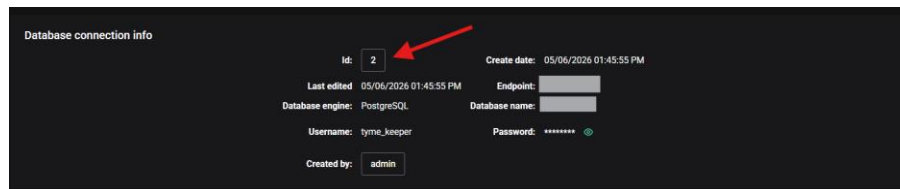
The header includes a sort by selection and a filter button to filter and sort your guards. Additionally, you can use the paginator on the bottom to change the view count.

The view-all webpage displays a summary of the most important information for the guard. You can select the guard’s name to view its details.

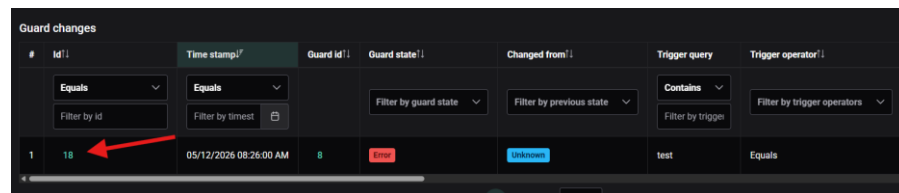


Selecting the guard’s name link navigates you to “/guards/detail/:id”. It displays The details of the guard, its database connection, its configured notifications, the guard state change history, and guard notification transactions sent. Many of the properties are links which can be used to navigate to other detail pages:

Link to database connection detail:



Link to guard change detail:



From the details page you can also manually run a guard using the “Run” button. Admins can also edit or delete guards here:

 Run  Edit guard  Delete guard

Guard change history

Understanding guard changes

When a guard state changes, a log is created for auditing purposes. The guard details at the time of change are stored in addition to the state it changed from and changed to. It also stores the result of the trigger column which caused the state change.

View guard changes

You can view guard changes by clicking the “Change history” navigation on the sidebar or navigating to “/guards/history-detail”.

This webpage displays a table of the guard state changes where you can sort, filter and navigate the paginated changes.

Clicking on the change id will navigate you to the change details at “/guards/history-detail/:id”.

Guard change history				
#	Id↑↓	Time stamp↑↓	Guard id↑↓	Guard state↑↓
	Equals Filter by id	Equals Filter by timesta	Equals Filter by guard id	Filter by guard state
1	18	05/12/2026 08:26:00 AM	8	Error
2	17	05/08/2026 09:56:20 AM	DELETED	Error

The change details will show a lot more information for the guard state change. Additionally, it will show the notifications that were sent out when that change occurred.

If the guard is not deleted it will also show the guards current information.

Database connections

Understanding a database connection

Create a database connection

Examples

Local database server

Remote database server

Local database file

- Endpoint: C:\Users\Leo\Downloads\database.db
- Database: SQLite
- Database Name: database.db

View database connections

You can view the database connections by clicking the “View connections” navigation on the sidebar. Alternatively, you can navigate to “/db-connections/view-all”.

This webpage will display a table with the database connection details. You can filter, sort and view the connection pages.

To view the details of the connection, you can click on the id link.

Database connections							
#	Id	Endpoint	Database engine	Username	Create date	Last edited	Created by
	Equals	Contains	Filter by database engine	Contains	Equals	Equals	Contains
	Filter by id	Filter by endpoint		Filter by username	Filter by create d	Filter by last edit	Filter by creat
1	4		SQLite		05/07/2026 04:25:08 PM	05/07/2026 04:25:08 PM	admin
2	3		SQLServer	test	05/07/2026 04:22:18 PM	05/07/2026 04:22:18 PM	admin
3	2		PostgreSQL	tyne_keeper	05/06/2026 01:45:55 PM	05/06/2026 01:45:55 PM	admin

The details can also be navigated to at “/db-connections/detail/:id”. It will display the database connection details, and the guards using this database connection.

From this page you can also test the database connection and administrators can edit or delete the connection.

+ DB Connection

Database connection info

Id: 4

Create date: 05/07/2026 04:25:08 PM

Last edited: 05/07/2026 04:25:08 PM

Endpoint:

Database engine: SQLite

Database name: db-test.db

Username:

Created by: admin

Check connection

Edit connection

Delete connection

Guards with database connection

#	Id	Guard Name	Created Date	Last edited	Last Run	Created By	Trigger Column	Trigger Operator	Trigger Val
	Equals	Contains	Equals	Equals	Equals	Contains	Contains	Filter by Trigger Operator	Equals
	Filter by id	Filter by Guard	Filter by	Filter by	Filter by	Filter by Creat	Filter by trigger		Filter by T

0 records across 0 pages

Notifications

Notification providers

Guard notifications

Understanding notifications

Create notifications

View notifications

Notification transactions

Understanding notification transactions

View notification transactions

You can view notification transaction by selecting “Notification transactions” from the sidebar menu. You can also navigate to “/guards/notification-transactions”.

The webpage will display a table of notification transactions which can be sorted, filtered and paginated.

You can select the transaction id link from the table to view the transaction details. This will navigate you to the details page at “/guards/notifications-transaction-detail/:id”.

This webpage will display the transaction information, and specific details for the type of transaction that was sent (Email or HTTP). It also will display the guard change information which triggered the notification.

Users

User info

Create users

View users

Miscellaneous

Dashboard

The dashboard is the landing page of the application. It lists a totals summary displaying the number of guards, notifications, providers, database connections, users and guard changes.

It also displays a helpful chart with the guard state changes for each month of the current year.

Settings

The settings webpage lists the current user's details and allows updating their username or password.

The user can also toggle light and dark modes here.